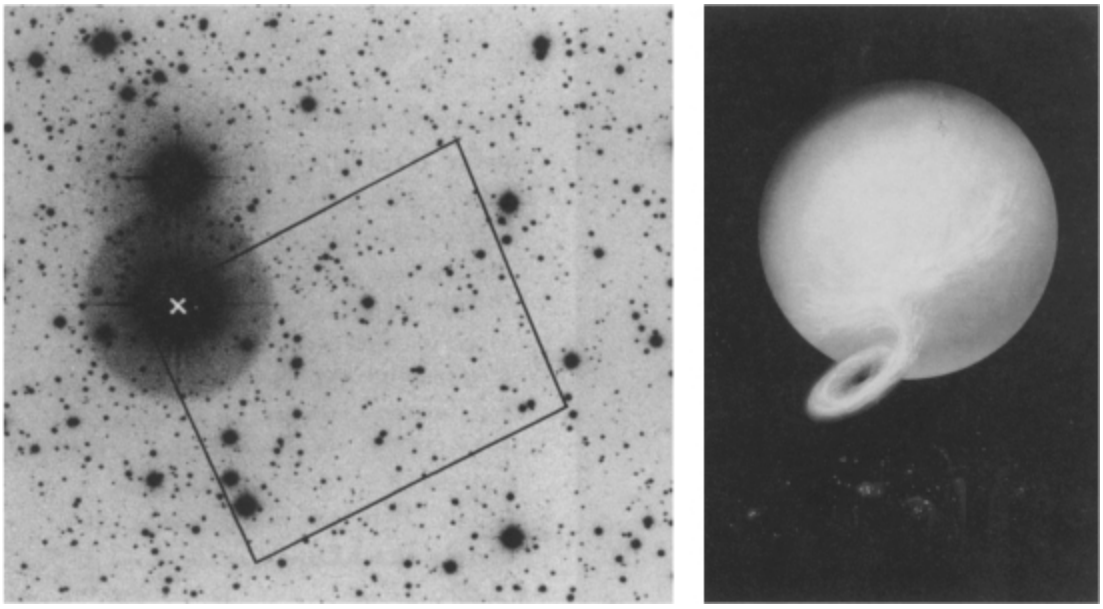
Stephen Murray, Ethan Schrier, and Riccardo Giacconi, who used Uhuru in 1971 to produce a 2-arc-minute-sized error box for the position of Cyg X-1 (Figure 8.7); and many others who discovered and studied violent, chaotic fluctuations of the X-rays and their energies—fluctuations that are what one would expect from hot, turbulent gas around a black hole.



8.7 Left: **A negative print of a photograph taken with the 5-meter (200-inch) optical telescope at Palomar Mountain by Jerome Kristian in 1971. The black rectangle outlines the error box in which Uhuru's 1971 data say that Cygnus X-1 lies. The white x marks the location of a radio flare, measured by radio telescopes, which coincided with a sudden change in the X-rays from Cyg X-1. The x coincides with the optical star HOE 226868, and thus identifies it as a binary companion of Cyg X-1. In 1978 the X-ray telescope Einstein confirmed this identification; see Figure 8.6g.** ***Right:*** **Artist's conception of Cyg X-1 and HDE 226868, based on all the optical and X-ray data.** [Left: photo courtesy Dr. Jerome Kristian, Carnegie Observatories; right: painting by Victor J. Kelley, courtesy the National Geographic Society.]

The observational astronomers contributing to the worldwide effort were people like Robert Hjellming, Cam Wade, Luc Braes, and George Miley, who discovered in 1971 a flare of radio waves in Uhuru's Cyg X-1 error box simultaneous with a huge, Uhuru-measured change in Cyg X-1's X-rays, and thereby pinned down the location of Cyg X-1 to within 1 second of arc (Figures 8.6d and 8.7); Louise Webster, Paul Murdin, and Charles Bolton, who discovered with optical telescopes that an optical star, HDE 226868, at the location of the radio flare is orbiting around a massive, optically dark but X-ray-bright companion (Cyg X-1); and a hundred or so other optical astronomers who made painstaking measurements of HDE